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iMotions Online

Facial Expression Analysis 2

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Facial Expression Analysis

This article describes the use of Facial Expression Analysis with iMotions Online. Affectiva 5.1 is currently used for the calculation of facial expressions. For more information on the AFFDEX algorithm, please click here.

Setting up your study

The collection of facial expression analysis data is done by default for all stimuli in iMotions Online. It cannot be modified on a stimulus-by-stimulus basis.

Viewing facial expression data

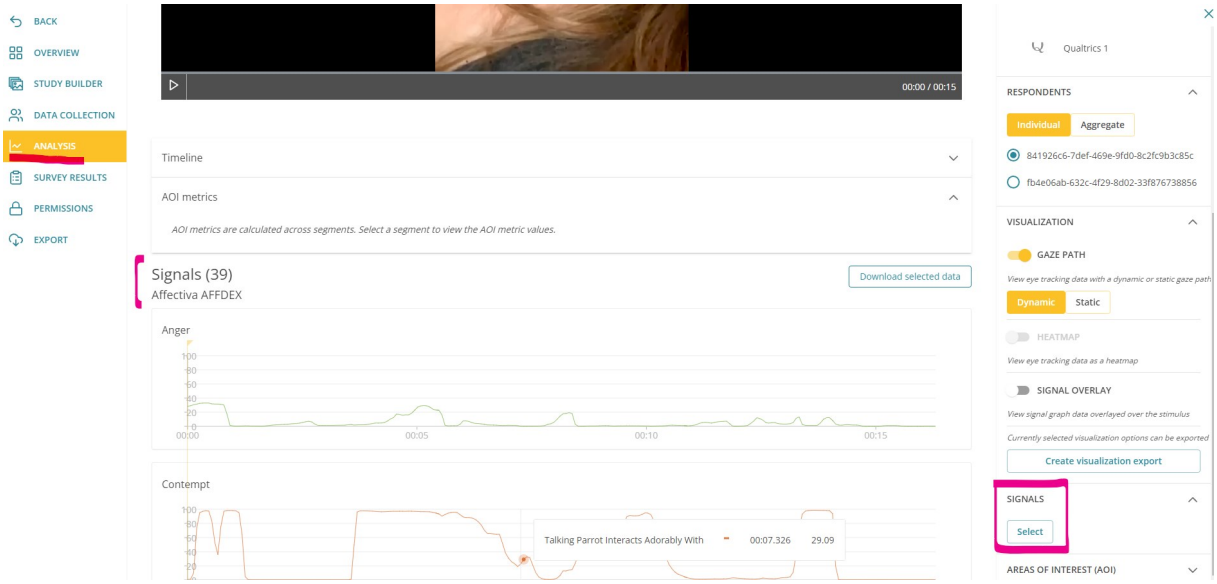
Note about facial expression videos

Please note that unlike iMotions Desktop (including remote data collection), respondent face videos are not available on the iMotions Online interface.

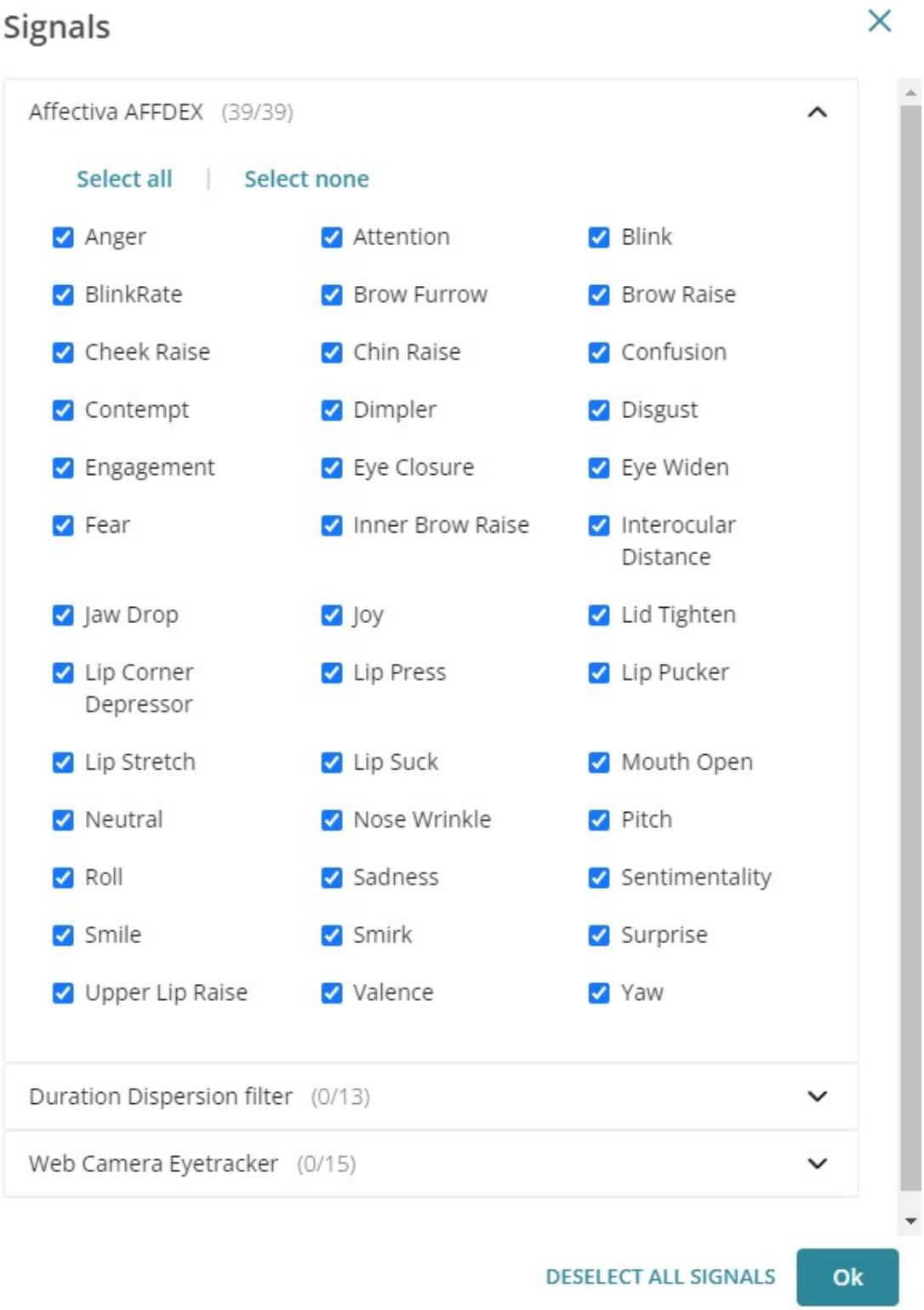
Facial expression data can be viewed by clicking the "Analysis" tab of your study. AFFDEX channels are shown in graphical format in the "Signals" section of the central panel, underneath your stimulus replay. By default, all graphs are made visible; to toggle on/off the graphs for the different action units and emotions, go to the "Signals" section of the right-hand menu, and click "Select".

Individual replays versus aggregate views can be selected from the "Respondents" section of the right-hand menu.

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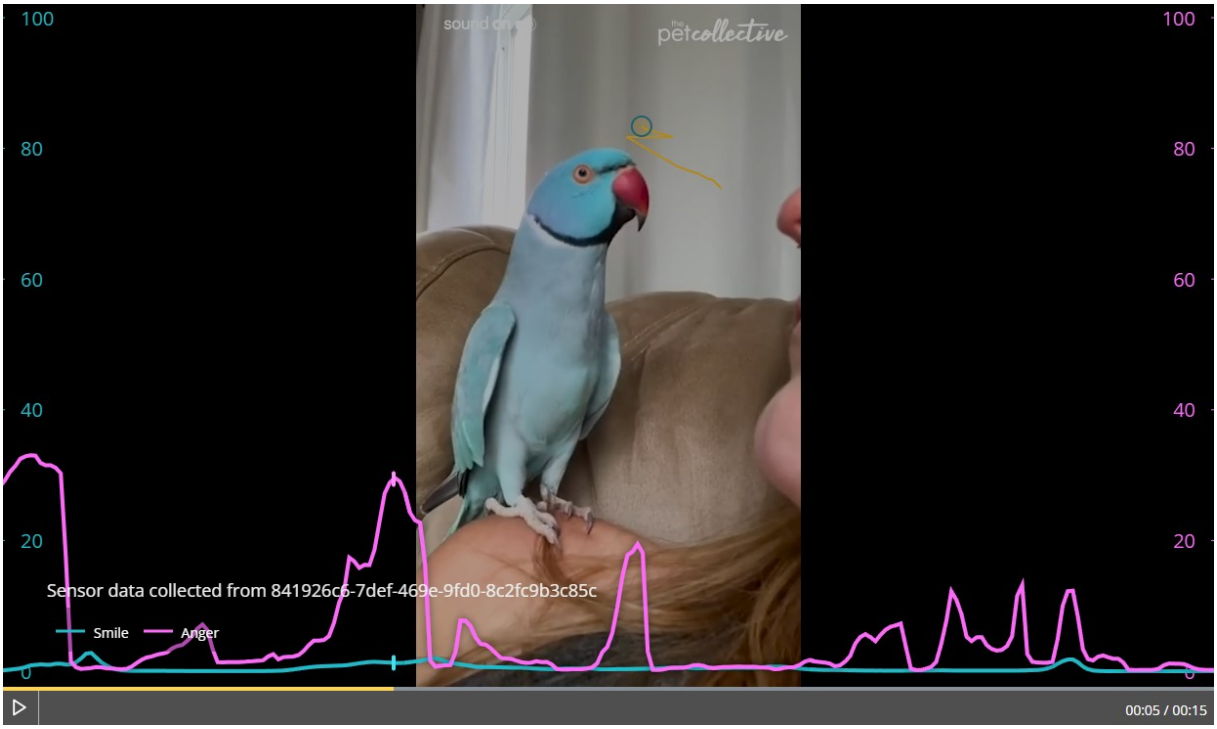
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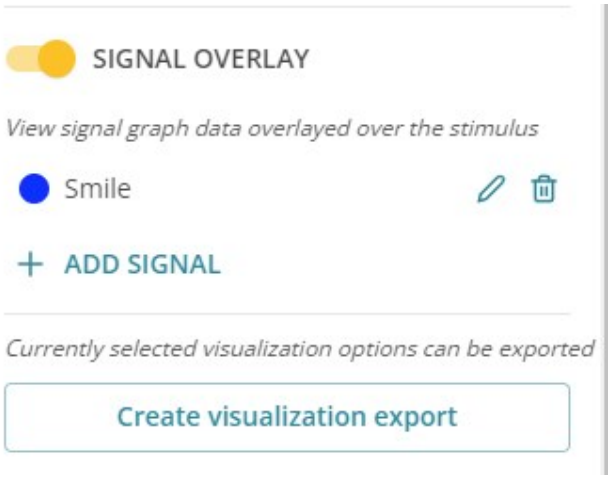
Alternate visualization: Signal overlay

In addition to the graphs described above, there is also the option to view facial expression data as an overlay on your stimulus, so it can be viewed synchronized with your eye-tracking data. Signal Overlay can be applied to all stimulus types.

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To enable this view, toggle "Signal Overlay" on the right-hand menu. You can then add the facial expression signals of your choice, and edit them for line width and color.



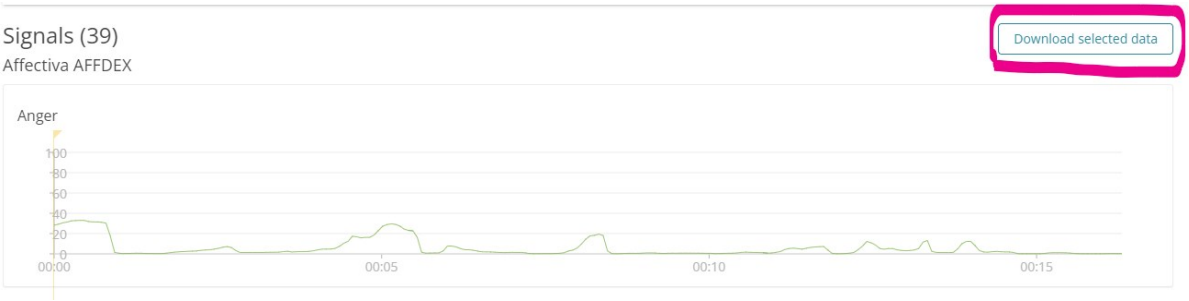
Signal overlay will only allow a maximum of two overlaid signals. You can also export the visualization with your chosen signal overlay.

Exporting facial expression data

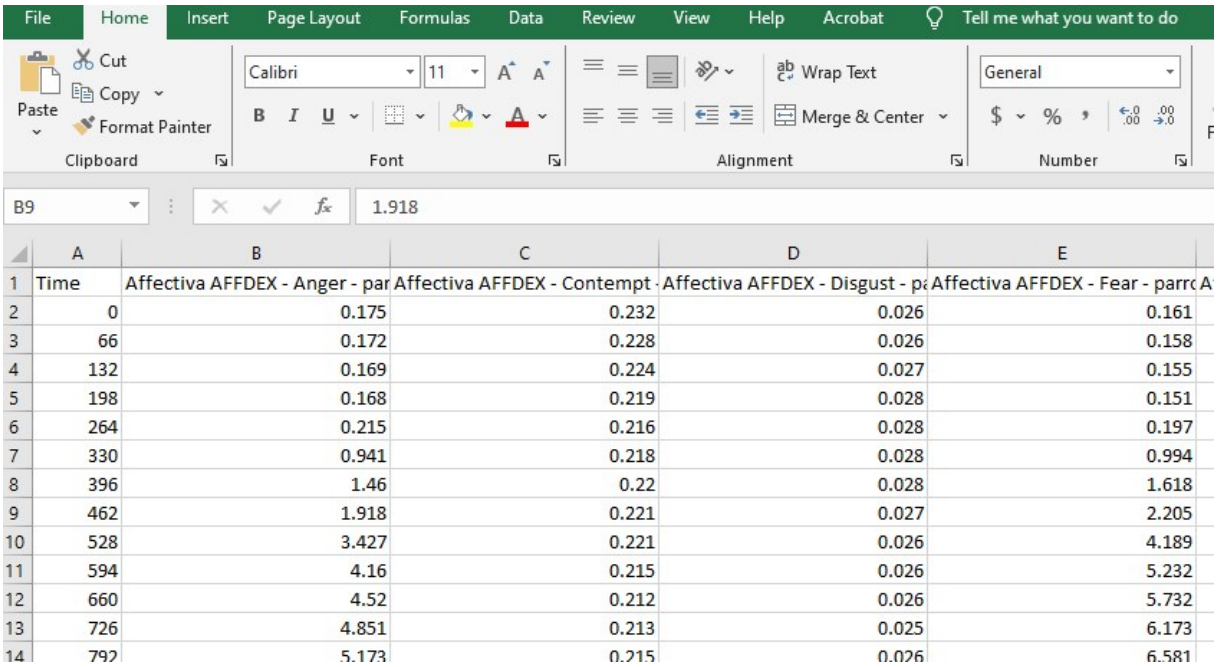
All of the facial expression graphs represent the raw data for AFFDEX, which is the probability that a trained human coder would rate that action unit/emotion as present at that time. Thus, all graphs are out of 100. Unlike iMotions Desktop, analysis tools like Signal Thresholding are currently not available, although you can toggle an aggregate view of the data across participants by selecting "Aggregate" in the Respondents section.

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It is possible to export the raw data from the signals graphs, by going to the Signals section in the central panel and clicking "Download selected data". This only downloads the data specific to that stimulus.



You can choose either US or European methods of handling decimals, and the resulting export is in .csv format.



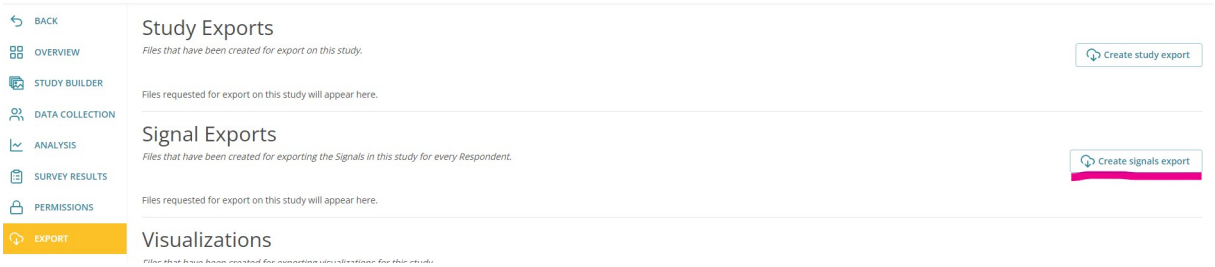
	A	B	C	D	E
1	Time	Affectiva AFFDEX - Anger - par	Affectiva AFFDEX - Contempt -	Affectiva AFFDEX - Disgust - pa	Affectiva AFFDEX - Fear - parrc
2	0	0.175	0.232	0.026	0.161
3	66	0.172	0.228	0.026	0.158
4	132	0.169	0.224	0.027	0.155
5	198	0.168	0.219	0.028	0.151
6	264	0.215	0.216	0.028	0.197
7	330	0.941	0.218	0.028	0.994
8	396	1.46	0.22	0.028	1.618
9	462	1.918	0.221	0.027	2.205
10	528	3.427	0.221	0.026	4.189
11	594	4.16	0.215	0.026	5.232
12	660	4.52	0.212	0.026	5.732
13	726	4.851	0.213	0.025	6.173
14	792	5.173	0.215	0.026	6.581

Warning: Only the signals that are made visible in the interface will be exported in text format.

Therefore, if you wish to export all of your data (including time-series data from eye-tracking), make sure all the graphs are toggled as visible before exporting. If you only care about one or two metrics, ensure those are the only graphs visible before exporting. This also includes whether data is showing individual respondents or aggregate view.

Exports can also be made en masse on the individual level by going to the "Export" tab and selecting "Create signals report".

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